FETCH Thesis Revisions: Chapters 1 and 2

Checked on 24 October 2026 against three things: the notes written on your PDF, the BarangayNet paper you gave as the reference, and the FETCH system as it is now built. Page numbers are those of "Jugan-Reusora_THESIS CHAP1&2.pdf" (44 pages).

How to use it: each change says where it goes and why. The text in the green boxes is ready to paste into your document; words shown in bold are the bold lead-ins your paper already uses. Keep your own page layout, font and spacing.

Summary of changes

No.	Page	Section	Change	Why
1	5, 8	Review of Related Literature	Add UAT and CSUQ	Note on p. 5: "Walang about sa csuq and uat"
2	9–10	Conceptual Diagram	New diagram and APA caption	Note on p. 9: "Change format"; reference paper
3	11	Hardware Requirements	Add internet; say what each part is for	The system is online
4	11	Software Requirements	Node.js, Python with scikit-learn, Render, Brevo	The system is built with these
5	12	Access Points	Three roles: Administrator, Cashier, Supplier	Note on p. 15; the system has three roles
6	12–13	Predictive Analytics & AI Module	Both algorithms, 30-day forecast	The system uses both
7	13	POS and Transaction Module	"Sensor" becomes barcode scanner	Matches the system
8	13	Reporting & Notifying Module	Emails to suppliers; "recharge" becomes "restock"	Matches the system
9	13	New: Supplier Management Module	Add the missing module	The diagram and objectives name it
10	14–15	Objectives of the Study	Unit testing checks; system testing by role	Note on p. 15; reference paper; your test plans
11	18–19	Scope and Delimitation	Java Spring Boot becomes Node.js; three roles; security	The system is Node.js
12	19–23	Definition of Terms	Add CSUQ, Purchase Order, Supplier Portal, System Testing, UAT, Unit Testing	Reference paper defines them
13	27	Development and Testing phases	Node.js; two levels of testing	Matches the system and test plans
14	28–29	Respondents of the Study	Exact numbers and the studies behind them	Notes on p. 28; reference paper
15	29	Guide Questions	Interview the owner/manager only, not suppliers	Note on p. 29

16	30	User Acceptance Testing (UAT)	Test case sheets per role; Node.js	Matches the test plans
17	30–31	Data Gathering Procedure	Both algorithms; unit and system testing	Matches the system
18	32	Statistical Treatment	Add the percentage formula	Reference paper
19	38–44	Appendix B	Replace the old unit testing plan	It no longer matches the tests

1. Review of Related Literature: add UAT and CSUQ (pages 5 and 8)

Why: The note on page 5 says the review has nothing about CSUQ and UAT, yet both are your evaluation tools (Objectives 2 and 3). The reference paper has a subsection for each (its pages 28 to 30). The studies cited below are the ones the reference paper uses; add them to your reference list (see the end of this document).

Where (a): Page 5, at the end of the first paragraph of Review of Related Studies and Literature ("…day-to-day operations of a pet supply store."). Add this sentence:

It also reviews User Acceptance Testing (UAT) and the Computer System Usability Questionnaire (CSUQ), the tools that will be used to test and evaluate the system.

Where (b): Page 8, after the paragraph Technology Acceptance and User Adoption in Retail Systems and before Pet Retail Industry. Add these two paragraphs:

User Acceptance Testing (UAT). User Acceptance Testing checks whether a system meets the needs of the people who will use it in their actual work. Pillai and Hemamalini (2022) showed that a hybrid UAT procedure, started early in development and repeated through the stages of the life cycle, detects defects earlier, makes debugging more efficient and lowers the risk of failure after deployment, compared with testing only at the end as in the Waterfall approach. Their study also found that UAT does more than show that a system works: it shows that it works for the people who will use it, which builds their trust in the system. FETCH applies this through two levels of UAT. In unit testing, technical experts test each module with completeness, format, range, consistency and database checks. In system testing, the end users of Risha Pet Supplies (the administrator, the cashier and the suppliers) carry out their own tasks screen by screen, from signing in to logging out.

Computer System Usability Questionnaire (CSUQ). The CSUQ is a standardized questionnaire that measures how usable a computer system is, as perceived by its users. Lewis (1995) developed it at IBM and reported coefficient alphas of 0.89 and above, showing that it is highly reliable. In a review of the questionnaires used to evaluate satisfaction, usability and acceptance in mobile health studies, Hajesmaeel-Gohari et al. (2022) found the CSUQ and its counterpart, the Post-Study System Usability Questionnaire (PSSUQ), among the most used instruments. FETCH uses the CSUQ to measure System Usefulness, Information Quality, Interface Quality and Overall Satisfaction from each type of user.

2. Conceptual Diagram: new diagram and caption format (pages 9 and 10)

Why: The note on page 9 says "Change format". The reference paper (its page 31) puts the figure number in bold on one line and the title in italics on the next (APA 7), and draws the schematic in black and white: requirements and methodology on top, then an arrow into the access points and the modules of each user, then the system name. The new diagram follows that layout and shows what FETCH really has, including the Cashier and Supplier modules the old one did not show.

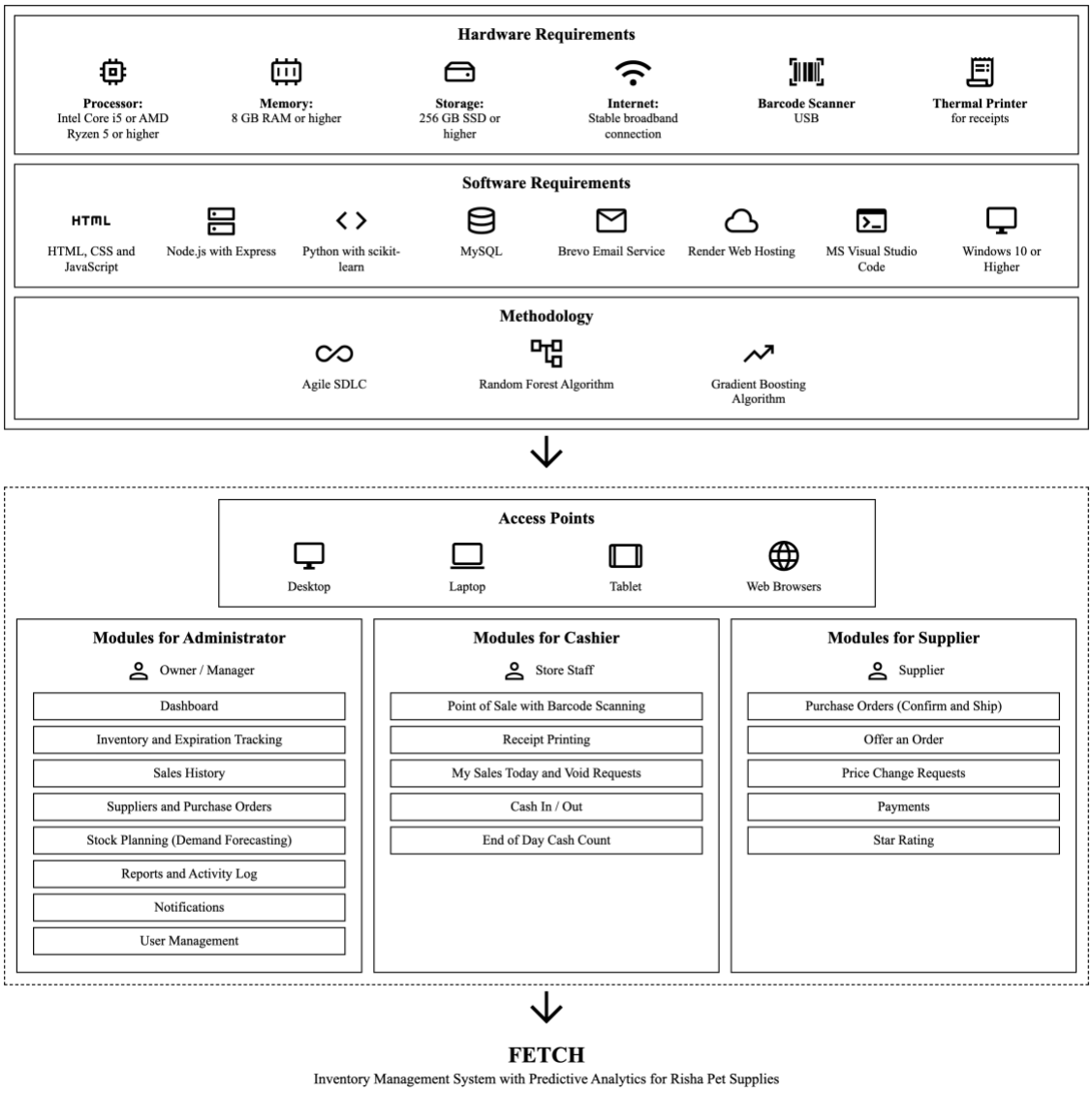
Replace: the paragraph under Conceptual Diagram with:

This section presents the schematic diagram as a visual representation of the FETCH system architecture. It shows how the hardware, software, methodology, access points and the modules of each user are designed to work together to achieve the objectives of the study.

Replace: "Figure 1. FETCH: Inventory Management System Schematic Diagram" with these two lines (first line bold, second line italic), then put the new diagram under them:

Figure 1

Schematic Diagram of FETCH



Add after the diagram (the reference paper explains its figure the same way):

Illustrated in Figure 1 is the architecture of FETCH, which is built to replace the manual inventory tracking of Risha Pet Supplies. The top part shows the hardware and software the system needs and the methodology used to build it: the Agile SDLC, and the Random Forest and Gradient Boosting algorithms that forecast demand. FETCH is a web-based system, so its users open it through a web browser on a desktop, laptop or tablet. Each user sees only the modules of their role. The administrator, who is the owner or manager, manages the inventory, sales, suppliers, purchase orders, users and reports, and uses Stock Planning to see the demand forecast

and what to reorder. The cashier, who is the store staff, uses the Point of Sale to scan products, complete sales, print receipts and count the cash at the end of the day. The supplier uses the Supplier Portal to confirm and ship purchase orders, offer orders, request price changes, and see its payments and star rating. Every sale, delivery and stock adjustment updates the same database, so all three users see the same figures.

3. Hardware Requirements (page 11)

Why: FETCH runs online, so it needs an internet connection, which the paragraph does not mention. The new text also says what each part is for.

Replace the paragraph with:

Hardware Requirements. The computer used to develop and run FETCH will require an Intel Core i5 or AMD Ryzen 5 processor or higher to handle the Random Forest and Gradient Boosting models, at least 8 GB of RAM so that the point of sale, the inventory pages and the forecast can run at the same time, and a 256 GB SSD or higher for fast access to the sales records. A stable internet connection will be needed, since FETCH is a web-based system that is opened through a web browser and sends emails to suppliers. A barcode scanner will be used to identify products at the point of sale, and a thermal printer will print the receipts.

4. Software Requirements (page 11)

Why: The paragraph does not name the back end. FETCH is built with Node.js and Express, the forecasting with Python and scikit-learn, it is hosted on Render and it sends email through Brevo. (Pages 18, 27 and 30 say Java Spring Boot; see changes 11, 13 and 16.)

Replace the paragraph with:

Software Requirements. FETCH will be developed on Windows 10 or 11 using Microsoft Visual Studio Code. The front end will use HTML, CSS and JavaScript to give the employees of the pet shop a clean and easy interface. The back end will use Node.js with the Express framework, and the demand forecasting will use Python with the scikit-learn library, which provides the Random Forest and Gradient Boosting algorithms. MySQL will be the main

database. FETCH will be hosted online on Render so that it can be opened from any web browser, and it will send emails to suppliers and users through the Brevo email service.

5. Access Points (page 12)

Why: The note on page 15 lists the three users: admin, supplier, cashier. The paragraph has only the Administrator and Staff access points; the Supplier is missing, and the Staff access point is the cashier's.

Replace the paragraph with:

Access Points. FETCH will be web-based and can be opened through a web browser on a desktop, laptop or tablet. To keep the data secure, each account will have one of three roles, and each role will see only its own pages. Through the Administrator Access Point, the store owner or manager will monitor all operations: the inventory, sales, suppliers, purchase orders, reports, the demand forecast and the user accounts. Through the Cashier Access Point, the store staff will go straight to the Point of Sale, a simple screen for scanning products, completing sales, recording cash taken in and out, and counting the cash at the end of the day. Through the Supplier Access Point, each supplier will see only its own purchase orders and products, where it can confirm and ship orders, offer an order, request a price change, and see its payments and star rating.

6. Predictive Analytics & AI Module (pages 12 and 13)

Why: It says the module uses the Random Forest algorithm only. FETCH uses Random Forest and Gradient Boosting together and forecasts the next 30 days.

Replace the paragraph with:

Predictive Analytics & AI Module. It is set to function as the backbone of the system. It will use two machine learning algorithms, Random Forest and Gradient Boosting, trained on the historical sales of the store, and combine their predictions to forecast how many of each product will sell in the next 30 days. It will learn patterns such as more of a particular brand of dog food

being bought on weekends or holidays, and use the forecast to tell the business owner which products to reorder, how many and by when.

7. Point of Sale (POS) and Transaction Module (page 13)

Why: "The system will incorporate a sensor" — the device is a barcode scanner. The first sentence ("The transaction module will facilitate this aspect.") also refers to nothing before it.

Replace the paragraph with:

Point of Sale (POS) and Transaction Module. The cashier will scan a product's barcode with a barcode scanner or choose it on the screen, and the system will complete the sale, print the receipt and subtract the items from the stock at once. The system will automatically stamp the date, the time and the cashier on each transaction, which forms the raw data for the reports and the demand forecast.

8. Reporting & Notifying Module (page 13)

Why: FETCH also emails purchase orders to the suppliers. "Recharge the inventory" should be "restock".

Replace the paragraph with:

Reporting & Notifying Module. The system will help the user keep the right amount of stock through visual reports and notifications. It will send "Low Stock" notifications once a product reaches its reorder level and will email purchase orders to the suppliers, giving the owner a chance to restock in time. The module will also give the management of Risha Pet Supplies detailed reports on sales trends to guide their decisions.

9. New paragraph: Supplier Management Module (page 13)

Why: The modules in the paper do not include supplier management, although the diagram, Objective 1 and the Significance of the Study (Suppliers) all name it.

Add after the Reporting & Notifying Module:

Supplier Management Module. The system will keep the record of each supplier and of the purchase orders sent to them. When the owner creates purchase orders from the reorder list, each supplier will receive its order by email and in the Supplier Portal, where it can confirm the

order, set the delivery date and mark it as shipped. Receiving an order will add the delivered items to the stock. Each supplier will also be given a star rating based on its delivery record, such as whether its orders arrive by the promised date.

10. Objectives of the Study (pages 14 and 15)

Why: The note on page 15 lists admin, supplier and cashier for system testing. The reference paper (its page 35) states who does each test: unit testing by technical experts and system testing by each type of end user. The checks are now the five in your Unit Test Plan; the screen-by-screen Functional Check became your System Test Plans, one for each role. Objective 1 also mentions "product locations", which FETCH does not track.

Replace Objective 1 with:

1. To develop FETCH: an Inventory Management System with Predictive Analytics for Risha Pet Supplies that implements ensemble machine learning algorithms (Random Forest and Gradient Boosting) to predict future demand based on historical sales patterns and seasonal trends, integrates barcode scanning technology for rapid product identification and automated inventory updates, applies an automated reordering mechanism that generates purchase orders for products at or below their reorder points, maintains real-time tracking of stock quantities and expiration dates, and provides a supplier portal to facilitate communication and coordination with suppliers.

Replace Objectives 2 and 3 with:

2. To test the system's functionality and acceptance through User Acceptance Testing (UAT), which will include:

2.1. Unit Testing by Technical Experts

2.1.1. Completeness Check

2.1.2. Format Check

2.1.3. Range Check

2.1.4. Consistency Check

2.1.5. Database Check

2.2. System Testing by End Users

2.2.1. Administrator (Store Owner or Manager)

2.2.2. Cashier (Store Staff)

2.2.3. Supplier

3. To evaluate the usability of the system using the Computer System Usability Questionnaire (CSUQ) in terms of:

3.1. System Usefulness (SYSUSE)

3.2. Information Quality (INFOQUAL)

3.3. Interface Quality (INTERQUAL)

3.4. Overall Satisfaction

11. Scope and Delimitation of the Study (pages 18 and 19)

Why: "Java Spring Boot as the backend technology" — FETCH is built with Node.js and Express. The scope also does not say FETCH is web-based or name its three users, and the security sentence leaves out the role-based access and email confirmation the system has.

Replace the last sentence of the first paragraph ("The system will be developed using Java Spring Boot…") with:

The system will be developed using Node.js with the Express framework as the back end, Python with the scikit-learn library for the machine learning models, MySQL as the database, and HTML, CSS and JavaScript as the front end. It will be a web-based system with three types of users: the administrator, the cashier and the supplier.

Replace the last sentence of the second paragraph ("Security will also include just basic user log-in…") with:

Security will include user log-in with role-based access, so that each user sees only the pages of their role, and email confirmation of new accounts; it will not include other advanced security features such as blockchain, biometrics and penetration testing.

12. Definition of Terms: add six terms (pages 19 to 23)

Why: The paper uses these terms but does not define them. The reference paper defines CSUQ, UAT and its system terms (its pages 40 to 42). Put each in its alphabetical place.

Computer System Usability Questionnaire (CSUQ). In this study, the CSUQ refers to the standardized questionnaire answered by the users of FETCH to rate its System Usefulness, Information Quality, Interface Quality and Overall Satisfaction (see Appendix A).

Purchase Order. Within this research, a purchase order is the order for stock that FETCH sends to a supplier, listing the products and quantities Risha Pet Supplies needs.

Supplier Portal. In this study, the Supplier Portal is the part of FETCH where a supplier signs in to see and confirm its purchase orders, mark them as shipped, offer orders, request price changes, and see its payments and star rating.

System Testing. For the purposes of this study, system testing is the test in which the administrator, the cashier and the suppliers each go through every screen of their role in order, from signing in to logging out, to check that FETCH works as a whole.

Unit Testing. For the purposes of this study, unit testing is the test in which technical experts check each module of FETCH separately through completeness, format, range, consistency and database checks.

User Acceptance Testing (UAT). In this study, UAT refers to the unit testing and system testing of FETCH, carried out to check that the system meets the needs of its users and is ready to be used at Risha Pet Supplies.

13. Agile phases: Development and Testing (page 27)

Why: Development says the back end is coded "with Java Spring Boot". Testing does not say who runs which test.

Replace the Development paragraph with:

Development. The system will be developed based on the design that has been accepted. This will entail coding the front end of the system with HTML, CSS and JavaScript, coding the back end with Node.js and the Express framework connected to a MySQL database, and coding the machine learning models with Python. The system will include functionalities such as inventory management, barcode scanning, automated reordering, supplier management, and inventory forecasting with the Random Forest and Gradient Boosting algorithms.

Replace the Testing paragraph with:

Testing. User Acceptance Testing (UAT) will be conducted in two levels to check that the system works as it should. In unit testing, technical experts will test each module using the Unit Test Plan, which covers completeness, format, range, consistency and database checks. In system testing, the administrator, the cashier and the suppliers will each follow the System Test Plan for their role, going through every screen from signing in to logging out. Afterwards, the users will answer the Computer System Usability Questionnaire (CSUQ) to evaluate the usability of the system.

14. Respondents of the Study (pages 28 and 29)

Why: The notes on page 28 say to give the highest number instead of a range ("Wag yung 5-10, specify na 10") and to base the section on the reference paper, which supports its sample sizes with studies (its pages 47 to 49). The numbers below take the top of each range: 10 suppliers, 8 technical experts and 10 pet shop owners or managers, 30 respondents in all.

Keep the first paragraph. Replace the second paragraph ("The respondents will include one (1) owner…") with these two paragraphs:

The sample size is based on established research in usability testing. Budiu (2021) explains that a qualitative usability test with five participants finds about 85% of the usability problems of an interface. For systems used by different types of users, Seinsche et al. (2023) showed that usability studies benefit from 8 to 15 participants per group, so that every type of user is represented. Caron et al. (2022) add that the sample size has to balance finding usability

problems against the resources available, and Sebele-Mpofu (2020) notes that purposive sampling of information-rich participants reaches data saturation without depending on a large sample.

Following these studies, the respondents will be one (1) owner and one (1) staff member of Risha Pet Supplies, who are the main users of the system as its administrator and cashier; ten (10) suppliers, who will test the Supplier Portal, including the purchase orders and notifications sent to them; eight (8) technical experts with a background in Computer Science, Information Technology or a related field, who will conduct the unit testing of each module and assess the system's architecture, functionality and the accuracy of its predictive analytics; and ten (10) owners or managers of other small pet shops, who will examine the system's relevance to the industry. In all, thirty (30) respondents will evaluate FETCH from operational, technical, business and supply chain perspectives. This approach makes the assessment thorough and fits the study's research methodology.

15. Guide Questions (page 29)

Why: The note on page 29 says the suppliers cannot be interviewed; only the admin or manager can.
Replace the paragraph with:

Guide Questions. The researchers will interview the owner and the manager of Risha Pet Supplies, who handle the inventory and the ordering from suppliers, to understand the requirements of the system. The interview will focus on how the store currently tracks its stock, how it decides what to reorder and when, and what the system needs to do to be useful in their daily work. The suppliers will not be interviewed; their part of the system will be based on the ordering process described by the owner, and they will take part only in the system testing and the CSUQ.

16. User Acceptance Testing (UAT) instrument (page 30)

Why: It says the back end is "built using Java Spring Boot", and it does not name the test case sheets. The reference paper describes its test case sheets as an instrument (its pages 50 and 51).

Replace the paragraph with:

User Acceptance Testing (UAT). The researchers will prepare test case sheets to check that FETCH works as expected. The Unit Test Plan will be used by the technical experts to test each module with completeness, format, range, consistency and database checks (see Appendix B). The System Test Plans, one each for the administrator, the cashier and the supplier, will be used by the end users to carry out their tasks screen by screen, such as adding a new product, scanning barcodes and completing sales at the point of sale, checking the demand forecast, creating purchase orders, confirming and shipping orders in the Supplier Portal, and tracking stock levels on the dashboard (see Appendices C to E). The tests will also check that the front end, the Node.js back end and the Python forecasting models work together correctly.

17. Data Gathering Procedure (pages 30 and 31)

Why: It names only the Random Forest algorithm, and the last paragraph describes unit testing as checking the forecast and alerts only.

Replace "…the main input for the training of the ensemble machine learning models, specifically the Random Forest algorithm." with:

…the main input for the training of the ensemble machine learning models, the Random Forest and Gradient Boosting algorithms.

Replace the last paragraph ("The researchers will finally collect evaluative data…") with:

The researchers will finally collect evaluative data through unit testing, system testing and the CSUQ. In unit testing, the technical experts will use the Unit Test Plan to check each module, including the accuracy of the stock forecast and the reorder alerts. As the modules are completed, the administrator, the cashier and the suppliers will follow the System Test Plan for their role and use the system as they would in their daily work. Their feedback will then be gathered through the CSUQ, which asks about the usability of the system and the reliability of

the AI-driven insights. This iterative collection of data will ensure that the final system actually minimizes costs and prevents stockouts as intended by the study.

18. Statistical Treatment: the percentage formula (page 32)

Why: The paper says a percentage will be calculated for each item but does not show how. The reference paper gives the formula (its page 54).

Add after the Percentage-Frequency Distribution paragraph:

Percentage. After the responses are tallied, each count will be turned into a percentage of all the responses, using the formula:

Percentage (%) = (Number of Responses in a Category ÷ Total Number of Responses) × 100

19. Appendix B: replace the old Unit Testing Plan (pages 38 to 44)

Why: The old plan says "Refer to UAT procedure" and "System behaves as expected" in every row, uses the word "Verify", and has a Functional Check and refund cases that no longer match your tests. The reference paper attaches its full test plans as appendices (its Appendices Z, AA and AB).

Replace Appendix B with the Unit Test Plan, and add the System Test Plans as new appendices, from your QA folder:

- **Appendix B:** Unit Test Plan (FETCH-Unit-Test-Plan-updated)
- **Appendix C:** System Test Plan, Administrator (FETCH-System-Test-Plan-Administrator)
- **Appendix D:** System Test Plan, Cashier (FETCH-System-Test-Plan-Cashier)
- **Appendix E:** System Test Plan, Supplier (FETCH-System-Test-Plan-Supplier)

References to add

These are the studies used in changes 1 and 14, copied from the reference list of the BarangayNet paper (APA 7). Your PDF has no reference list; add them to wherever your references are kept.

Budiu, R. (2021, July 11). Why 5 participants are okay in a qualitative study, but not in a quantitative one. Nielsen Norman Group. https://www.nngroup.com/articles/5-test-users-qual-quant/

Caron, A., Vandewalle, V., Marcilly, R., Rochat, J., & Dervaux, B. (2022). The optimal sample size for usability testing, from the manufacturer's perspective: A value-of-information approach. Value in Health, 25(1), 116–124. https://doi.org/10.1016/j.jval.2021.07.010

Hajesmaeel-Gohari, S., Khordastan, F., Fatehi, F., Samzadeh, H., & Bahaadinbeigy, K. (2022). The most used questionnaires for evaluating satisfaction, usability, acceptance, and quality outcomes of mobile health. BMC Medical Informatics and Decision Making, 22, 22. https://doi.org/10.1186/s12911-022-01764-2

Lewis, J. R. (1995). IBM computer usability satisfaction questionnaires: Psychometric evaluation and instructions for use. International Journal of Human-Computer Interaction, 7(1), 57–78. https://doi.org/10.1080/10447319509526110

Pillai, R., & Hemamalini, R. (2022). Hybrid user acceptance test procedure to improve the software quality. The International Arab Journal of Information Technology, 19(6), 947–954. https://www.iajit.org/portal/images/Year2022/No.6/20887.pdf

Sebele-Mpofu, F. Y. (2020). Saturation controversy in qualitative research: Complexities and underlying assumptions. A literature review. Cogent Social Sciences, 6(1), Article 1838706. https://doi.org/10.1080/23311886.2020.1838706

Seinsche, J., de Bruin, E. D., Saibene, E., Rizzo, F., Carpinella, I., Ferrarin, M., Moza, S., Ritter, T., & Giannouli, E. (2023). A newly developed exergame-based telerehabilitation system for older adults: Usability and technology acceptance study. JMIR Human Factors, 10, Article e48845. https://doi.org/10.2196/48845

Not changed, but worth a look before the defense

- **Page 6:** "The FETCH system focuses on achieving this validated accuracy level of 94.3%." The forecast page measured 86.1% for 27 Sep to 10 Oct (QA/FORECAST-METHOD.md). The panel may ask, so consider: "FETCH aims for an accuracy close to this level."
- **Page 7:** "the automatic reorder system places automatic orders". In FETCH the owner reviews the reorder list and creates the purchase orders with one click; consider "prepares purchase orders that the owner approves".
- **Page 9:** "ScanBarcode" reads like a product name; "barcode scanning" is clearer.
- **Pages 2 to 4 (Introduction):** the tone is casual for a thesis ("doing really well now", "Think about what it takes"), and "Intelligence and machine learning" should be "Artificial intelligence and machine learning".
- **Tense:** the paper says the system "will" be built, but the consent form says it is already live. The reference paper uses the past tense in Chapter 2 for work already done. Ask your adviser which they want.